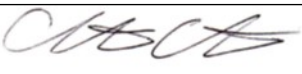


Guideline Title: Intravenous (IV) Fluid Therapy

Guideline Number: GUID-3203-PR015

Issue date: 09/09/2014	Replaces Dept. Policy: IV Fluid Therapy
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To provide guidelines for fluid resuscitation and maintenance of fluid administration
- II. **DEFINITIONS:** IV - Intravenous
- III. **DEPARTMENT GUIDELINE:**
 - a. **EQUIPMENT:** IV Fluids: 0.9% Normal Saline (NS), Lactated Ringers, Dextrose containing solutions
 - b. **INDICATIONS:** Hemodynamic instability requiring fluid replacement or medications
 - c. **PROCEDURE: General**
 - i. All patients will have an IV line/saline lock in place unless contraindicated.
 - ii. Pt's with hypovolemia, potential for hypovolemia, or hemodynamic instability present should preferably have two large bore IVs.
 - iii. If unable to establish a peripheral IV line, proceed to intraosseous (IO) infusion.
 - iv. IV fluid replacement is based on patient condition and appropriate standard of care.
 - a. If hypotension present, treat per GUID3203-M011 – Hypotension or per GUID3203-PED010 – Pediatric Hypotension.
 - b. Limit IV fluids in trauma patients. If active hemorrhage, treat with blood products per GUID3203-PRO25 – Emergency Blood Product Administration.
 - c. For burn patients treat per GUID3203-T002 – Burns.
 - v. For adults, utilize pressure bag for rapid infusion if indicated.
 - d. **PROCEDURE: Pediatric**
 - i. If delivering an IV bolus, consider 'push-pull' method using a syringe attached to a fluid bag with a 3-way stopcock to draw IV fluid into syringe then deliver to patient using repetitive strokes of the syringe.
 - ii. Following acute resuscitation, consider a dextrose containing maintenance fluid infusion in addition to other volume replacement (e.g. septic patients, neonates, burn patients).
 1. Maintenance rate:
 - a. Using the 4-2-1 rule
 - i. 4ml/kg/hr for the 1st 10kg of body weight *Plus*
 - ii. 2ml/kg/hr for the 2nd 10kg of body weight *Plus*
 - iii. 1ml/kg/hr for each additional kg of body weight

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b. Alternate formula:

- i. 0-10 kg --- $100 \text{ ml/kg} \div 24 \text{ hrs} = \text{hourly rate}$
- ii. 0-20 kg --- $1000 \text{ ml} + 50 \text{ ml/kg for each kg} > 10 \text{ kg} \div 24 \text{ hrs} = \text{hourly rate}$
- iii. 20 kg or > --- $1500 \text{ ml} + 20 \text{ ml/kg for each kg} > 20 \text{ kg} \div 24 \text{ hrs} = \text{hourly rate}$

IV. **DOCUMENTATION:** EMS Charts

V. **REFERENCES:** None